Advanced Programming Practice · 21CSC203P

Unit 4 · Session 6 · SLO 2 — Solved Worksheet

Name: R L KRISHE YADAV R L KRISHE YADAV | **Reg. No:** RA2511030010473

Unit â€“ 4 : Session â€“ 6 : SLO â€“ 2 | Parallel Computing â€” Architecture Matching

I. Match the Following

1. Shared Memory â€” Common address space: all processors read/write one global memory.
2. Distributed Memory â€” Message passing: each node has private memory; data moves via explicit send/receive (MPI).
3. Multicore CPU â€— Parallel execution units: several cores on one chip execute threads simultaneously.

II. Key Notes

• Flynn's taxonomy context: multicore CPUs and shared-memory machines are typically MIMD systems.
• Hybrid systems combine both: clusters of multicore nodes use shared memory inside a node and message passing across nodes.